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🧠 Enhanced Email Classification Model Trainer
   Version 2.0 - Advanced ML Techniques
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📂 Loading dataset...
   Total samples: 315
   Label distribution:
      phishing: 179
      legitimate: 85
      spam: 51

🔧 Advanced preprocessing...
📊 Extracting additional features...

📊 Data split:
   Training samples: 252
   Testing samples: 63

🔤 Enhanced TF-IDF Vectorization...
   Vocabulary size: 570
   Feature matrix shape: (252, 570)

🎯 Training enhanced models with cross-validation...

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📊 Naive Bayes (Tuned)
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   CV F1-Scores: [0.88030944 0.89848923 0.87866667 0.90083089 0.86206061]
   Mean CV F1: 0.8841 (+/- 0.0285)
   Test Accuracy:  0.7143
   Test Precision: 0.7025
   Test Recall:    0.7143
   Test F1-Score:  0.6966

📊 Logistic Regression (Optimized)
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619
   Test F1-Score:  0.7487
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📊 Random Forest (Enhanced)

CV F1-Scores: [0.80883713 0.81682479 0.81325653 0.81409524 0.88242424]
Mean CV F1: 0.8271 (+/- 0.0556)
Test Accuracy: 0.8095
Test Precision: 0.8077
Test Recall: 0.8095
Test F1-Score: 0.8083

📊 Gradient Boosting

CV F1-Scores: [0.860181 0.81543333 0.76384773 0.71082717 0.68853002]
Mean CV F1: 0.7678 (+/- 0.1276)
Test Accuracy: 0.7619
Test Precision: 0.7830
Test Recall: 0.7619
Test F1-Score: 0.7556

📊 Ensemble (Voting)

CV F1-Scores: [0.92281065 0.92042484 0.89602886 0.8982682 0.83873684]
Mean CV F1: 0.8953 (+/- 0.0606)
Test Accuracy: 0.8095
Test Precision: 0.8023
Test Recall: 0.8095
Test F1-Score: 0.7968

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🏆 Best Model: Ensemble (Voting)
Combined Score: 0.8460

📋 Detailed Classification Report for Ensemble (Voting):
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	precision	recall	f1-score	support
Legitimate	0.87	0.76	0.81	17
Spam	0.67	0.40	0.50	10
Phishing	0.81	0.94	0.87	36
accuracy			0.81	63
macro avg	0.78	0.70	0.73	63
weighted avg	0.80	0.81	0.80	63

📊 Confusion Matrix:
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[[13  0  4]
 [ 2  4  4]
 [ 0  2 34]]
```

📈 Per-Class Accuracy:
  Legitimate: 0.7647
  Spam: 0.4000
  Phishing: 0.9444

💾 Model saved to: C:\Users\sahil\OneDrive\Desktop\SPAM EMAIL DETECTION\models\email_classifier.pkl
💾 Vectorizer saved to: C:\Users\sahil\OneDrive\Desktop\SPAM EMAIL DETECTION\models\tfidf_vectorizer.pkl

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📊 TRAINING SUMMARY
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Model	Accuracy	F1-Score	CV Mean
Random Forest (Enhanced)	0.8095	0.8083	0.8271
Ensemble (Voting)	0.8095	0.7968	0.8953
Logistic Regression (Optimized)	0.7937	0.7838	0.8793
Gradient Boosting	0.7619	0.7556	0.7678
SVM (Calibrated)	0.7619	0.7487	0.8833
Naive Bayes (Tuned)	0.7143	0.6966	0.8841

✅ Enhanced training complete!
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